

Final Narrative – Decision Deck

Subscription Churn Early-Warning & Retention Strategy

Slide 1 — Title Slide

Title:

Subscription Churn Early-Warning & Retention Strategy

Subtitle:

Data-Driven Retention Plan for a Subscription Platform

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Date: 10/03/2026

Slide 2 — Goal, KPIs & Churn Definition

Business Goal:

Reduce subscriber churn by identifying **early churn signals** and implementing **targeted retention interventions**.

North Star Metric:

Monthly Retention Rate

Percentage of users who remain subscribed during the month.

Supporting KPIs:

- Churn Rate (weekly/monthly)
- Cohort Retention
- Renewal Success Rate
- Payment Failure Rate
- Average Usage Days per Week
- Revenue Retained / Revenue Lost
- Campaign Response Rate

Churn Definition

A user is considered **churned if:**

Subscription is cancelled **OR** not renewed within **7 days after subscription expiry**.

Slide 3 — Data & Methodology

Data Sources

Six datasets were used:

- Users dataset (user profiles)
- Subscription lifecycle data
- Plans and pricing
- Payment attempts and outcomes
- Daily product usage
- Campaign touchpoints

Data Processing (ETL Pipeline)

Python ETL pipeline was developed to:

- Clean and standardize raw data
- Remove duplicates
- Normalize plan names and payment status
- Handle missing values
- Flag outliers

Curated Datasets Created

1. *dim_users_enriched*
 - User demographics
 - Tenure
 - Engagement band
2. *fact_user_weekly*
 - Weekly usage metrics
 - Payment attempts
 - Renewal signals
3. *model_churn_dataset*
 - ML features from last 4 weeks
 - churn prediction label (will_churn_14d)

Slide 4 — Key Insights

The analysis revealed *five major churn drivers*.

1. Engagement Drop Before Churn
Users who churned showed a **40–45% decline in usage** in the final weeks.
2. Early Tenure Churn
New users (<30 days) had the **highest churn rate** due to weak onboarding.
3. Plan Type Differences

Monthly plan users churn **3× more** than annual plan users.

4. Payment Failures

Over *40% of churners experienced payment failures* before cancellation.

5. Low Engagement Segments

Users in the *low engagement band* have the highest churn risk.

Slide 5 — Segment Deep Dive

Key high-risk segments identified:

Highest Churn Segments

1. Monthly plan + low engagement users
2. New users (tenure < 30 days)
3. Users with recent payment failures

Highest Revenue-at-Risk Segments

1. Monthly plan users
2. Mid-tenure users (30–90 days)
3. Tier-2 city users

These segments will be *prioritized in the retention strategy*.

Slide 6 — Churn Prediction Model

A machine learning model was built to *predict churn risk 14 days in advance*.

Key Model Features

- Usage trend (last week vs previous weeks)
- Active days per week
- Payment failures
- Days since last activity
- Plan type and tenure.

Model Output

Users are segmented into **risk bands**:

Risk Band	Score
High Risk	>0.70
Medium Risk	0.40–0.70
Low Risk	<0.40

Targeting Strategy

Retention interventions will prioritize:

- High-risk users
- Medium-risk users with declining usage

Slide 7 — Retention Strategy

- A targeted retention program is proposed.

Targeting Logic

Users will be targeted based on:

- Churn risk score
- Engagement decline
- Renewal proximity

Key Interventions

1. Renewal Reminder
Sent **7 days before renewal**
2. Personalized Content Recommendations
Suggested features or content to increase engagement.
3. Limited-Time Retention Offers
Short-term discounts for high-risk users.
4. Payment Retry Nudges
Notifications to update payment method after failures.

Slide 8 — Experiment Design

Retention strategies will be validated through **A/B testing**.

Experiment Setup

Group	Description
Control	No intervention
Treatment	Receives retention campaign

Primary Metric

Renewal / retention rate.

Guardrail Metrics

- Discount cost
- Revenue margin
- Complaint rate

- Engagement improvement

Experiment Duration

3–4 weeks to cover a full billing cycle.

Slide 9 — 30-Day Impact Projection

If the retention strategy is implemented:

Scenario	Users Saved	Revenue Retained
Worst Case	600	₹3L
Base Case	1,440	₹7.2L
Best Case	2,400	₹12L

Estimated Incentive Cost

₹4.5L (discount offers)

Net Impact

Base scenario:

₹2.7L net revenue gain.

Slide 10 — Risks, Limitations & Next Steps

Risks / Limitations

- Prediction accuracy may vary across segments.
- Limited campaign data for response analysis.
- Short historical window for some users.

